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VIDEO CAPTURE DEVICE CONTROL BASED ON METADATA RELATED TO A VIDEO ENVIRONMENT

Abstract

Techniques are disclosed herein for providing video capture device control based on metadata related to a video environment. Examples may include receiving metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment, generating a view quality score for the video environment based at least in part on the metadata, generating control data for the at least one video capture device based at least in part on the view quality score, and outputting the control data to the at least one video capture device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/556,431, titled "VIDEO CAPTURE DEVICE CONTROL BASED ON METADATA RELATED TO A VIDEO ENVIRONMENT," and filed on Feb. 22, 2024, the entirety of which is hereby incorporated by reference.

TECHNICAL FIELD
[0002] Embodiments of the present disclosure relate generally to video processing and, more particularly, to systems configured to control video capture devices.
BACKGROUND
[0003] Applicant has identified many deficiencies and problems associated with existing techniques for capturing, processing, and/or transmitting video data captured by video captured devices in a video environment. Through applied effort, ingenuity, and innovation, many of these identified deficiencies and problems have been solved by developing solutions that are configured in accordance with embodiments of the present disclosure, many examples of which are described herein.
BRIEF SUMMARY
[0004] Various embodiments of the present disclosure are directed to apparatuses, systems, methods, and computer readable media for providing video capture device control based on metadata related to a video environment. These characteristics as well as additional features, functions, and details of various embodiments are described below. The claims set forth herein further serve as a summary of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] Having thus described some embodiments in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:
[0006] FIG. **1** illustrates an example video processing system configured to execute audio/video (AV) processing operations related to video events in accordance with one or more embodiments disclosed herein;
[0007] FIG. **2** illustrates an example AV processing apparatus configured in accordance with one or more embodiments disclosed herein;
[0008] FIG. **3** illustrates an example network system in accordance with one or more embodiments disclosed herein;
[0009] FIG. **4** illustrates an example transmitter system in accordance with one or more embodiments disclosed herein;
[0010] FIG. **5** illustrates an example receiver system in accordance with one or more embodiments disclosed herein;
[0011] FIG. **6** illustrates an example score computation architecture in accordance with one or more embodiments disclosed herein;
[0012] FIG. **7** illustrates an example device control architecture in accordance with one or more embodiments disclosed herein;

[0013] FIG. **8** illustrates an example machine learning architecture in accordance with one or more embodiments disclosed herein;
[0014] FIG. **9** illustrates an example video environment in accordance with one or more embodiments disclosed herein;
[0015] FIG. **10** illustrates an example method for providing video capture device control based on metadata related to a video environment in accordance with one or more embodiments disclosed herein; and
[0016] FIG. **11** illustrates another example method for providing video capture device control based on metadata related to a video environment in accordance with one or more embodiments disclosed herein.

DETAILED DESCRIPTION

[0017] Various embodiments of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the present disclosure are shown. Indeed, the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements.

Overview

[0018] An audio video (AV) conferencing system may include multiple video cameras to capture video data in a video environment. The captured video data may be transmitted between devices in the video environment and/or another environment via a network. For example, a remote hub may receive and process the captured video data from the video cameras. The remote hub may also transmit the processed video data to one or more display devices in the video environment and/or another environment via a network. In certain scenarios, a video environment may be a conference room environment with multiple video cameras. However, in such a scenario, one or more of the video cameras may transmit unnecessary and/or irrelevant video streams for the video environment, resulting in inefficient bandwidth usage of video transmission for the video environment. As such, in certain scenarios such as a conference room environment with multiple video cameras, it may be desirable to utilize a subset of the video cameras to transmit live video feeds due to the high bandwidth usage of video transmission.

[0019] To address these and/or other technical problems associated with traditional AV conferencing systems, various embodiments disclosed herein provide video capture device control based on metadata related to a video environment. For example, one or more machine learning models, one or more digital signal processing (DSP) techniques, and/or one or more other prediction techniques configured to derive characteristics for a video environment may be utilized to generate metadata of video streams. In various examples, the metadata may include machine learning model inference results, DSP insight results, and/or other information related to the video environment. The metadata may be based on video and/or audio related to the video streams. Additionally, the metadata may be transmitted over a network to allow enabling or disabling of respective video capture devices within a video environment. In some examples, the machine learning models may be deployed on the video capture devices and the metadata provided by the machine learning models is transmitted to a hub device that utilizes the metadata to intelligently select which video capture devices to enable in a video environment. The metadata may be additionally or alternatively utilized to intelligently configure the respective video capture devices to, for example, focus on a target of interest (e.g., determine where and/or how to “point” a video capture device to capture the target of interest) in the video environment.

[0020] Accordingly, certain machine learning tasks may be offloaded to edge devices while at the same time mitigating inefficient transmission of irrelevant video streams in the video environment. Additionally, based on the metadata, real-time selection and/or composition of video capture devices in a video environment may be provided. Moreover, the metadata may be utilized to optimize video and/or audio dynamics for a video environment to provide a media consumption experience that provides equity for both remote and in-person attendees of the video environment. By utilizing the metadata as disclosed herein, network latency and/or bandwidth utilization for transmitting video data may also be minimized. Additionally, efficiency and/or quality of video processing by a video capture device may be additionally or alternatively improved.

Example Video Processing Systems and Methods

[0021] FIG. **1** illustrates a video processing system **100** that is configured to provide a multi-threaded video pipeline for video content related to a video environment, according to embodiments of the present disclosure. For example, the video processing system **100** provides real-time configuration of a video capture devices in a video environment. The video processing system **100** may be, for example, a video environment system, a conferencing system (e.g., a conference audio system, a video conferencing system, an audio video (AV) conferencing system, a digital conference system, etc.), a lecture hall system, a classroom system, a live event system, an automobile advanced driver assistance system (ADAS), a digital media content workstation, a broadcasting system, an augmented reality system, a virtual reality system, a gaming system, an online gaming system, or another type of video system. Additionally, the video processing system **100** may be implemented as a video processing apparatus and/or as software that is configured for execution on a network device, a video capture device (e.g., a camera device), a smartphone, a laptop, a personal computer, a digital conference system, a wireless conference unit, a video workstation device, a communication center device (e.g., a hub device), or another device. The video processing system **100** disclosed herein may additionally or alternatively be integrated into a virtual video processing system (e.g., video processing via virtual processors or virtual machines) with other audio and/or digital signal processing.

[0022] By providing real-time configuration of a video capture devices, the video processing system **100** may provide various improvements related to video processing such as, for example: minimizing network latency for transmitting video data over a network, minimizing bandwidth utilization for transmitting video data over a network, reducing a number of computing resources for processing video by a video capture device, and/or improving power consumption for processing video by a video capture device. The video processing system **100** may also be adapted to produce improved video signals for a video environment. Additionally or alternatively, the video processing system **100** may be adapted to produce improved audio for video signals. For example, audio for video signals may be provided with reduced noise, reduced reverberation, improved source separation, and/or a reduction in other undesirable audio artifacts. A video environment may be an indoor environment, an outdoor environment, an entertainment environment, a room, a conference room, a meeting room, a classroom, a lecture hall, a performance hall, a broadcasting environment, a sports stadium or arena, a virtual environment, an automobile environment, or another type of video environment.

[0023] In some examples, the video processing system **100** may provide video streams for rendering via one or more display devices. In some examples, a display device may receive a video stream via a physical interface protocol such as a universal serial bus (USB) communication protocols or another type of communication protocol. In some examples, a display device may receive a video stream via a network communication protocol such as an Internet Protocol (IP), IP over Ethernet (IPoE), or other network communication protocol. In some examples, a display device may be a virtual camera or another type of virtual device.

[0024] The video processing system **100** includes one or more video capture devices **103**. The one or more video capture devices **103** may respectively be devices configured to capture video related to the one or more sound sources and/or one or more entity of interest in a video environment. A sound source may be, for example, a speaker in a conference room. An entity of interest may be, for example, a whiteboard surface or a non-speaking person in a conference room. The one or more video capture devices **103** may include one or more sensors configured for capturing video by converting light into one or more electrical signals. The video captured by the one or more video capture devices **103** may also be converted into video data **105**. In an example, the one or more video capture devices **103** are one or more video cameras.

[0025] In some examples, the video processing system **100** additionally includes one or more audio capture devices **102**. The one or more audio capture devices **102** may respectively be devices configured to capture audio from one or more sound sources. The one or more audio capture devices **102** may include one or more sensors configured for capturing audio by converting sound into one or more electrical signals. The audio captured by the one or more audio capture devices **102** may also be converted into audio data **106**. The audio data **106** may be a digital audio data or, alternatively, analog audio data, related to the one or more electrical signals. In some examples, the audio data **106** may be beamformed audio data.

[0026] In an example, the one or more audio capture devices **102** are one or more microphones arrays. For example, the one or more audio capture devices **102** may correspond to one or more array microphones, one or more beamformed lobes of an array microphone, one or more linear array microphones, one or more ceiling array microphones, one or more table array microphones, or another type of array microphone. In alternate examples, the one or more audio capture devices **102** are another type of capture device such as, but not limited to, one or more condenser microphones, one or more micro-electromechanical systems (MEMS) microphones, one or more dynamic microphones, one or more piezoelectric microphones, one or more virtual microphones, one or more network microphones, one or more ribbon microphones, and/or another type of microphone configured to capture audio. It is to be appreciated that, in certain examples, the one or more audio capture devices **102** may additionally or alternatively include one or more infrared capture devices, one or more sensor devices, one or more video capture devices (e.g., one or more video capture devices **103**), and/or one or more other types of audio capture devices.

[0027] The one or more video capture devices **103** and/or the one or more audio capture devices **102** may be positioned within a particular video environment. In some examples, the video data **105** includes video frames related to a speaker associated with the audio data **106**. In some examples, the one or more video capture devices **103** and the one or more audio capture devices **102** may be integrated together in one or more capture devices.

[0028] The video processing system **100** also comprises an audio/video (AV) processing system **104**. The AV processing system **104** may be configured to perform one or more video processes and/or one or more audio processes with respect to the video data **105** and/or the audio data **106** to provide encoded video data **114**. The AV processing system **104** depicted in FIG. **1** includes a video scoring engine **109**, device control engine **110**, a video pipeline engine **111**, and/or an audio pipeline engine **112**.

[0029] The video scoring engine **109** utilizes metadata **101** provided by one or more machine learning models **120** and/or a metadata engine **122** to generate a view quality score **107** related to the one or more video capture devices **103**. The view quality score may refer to a numerical or categorical value that represents quality of a view captured by the one or more video capture device **103**. The quality of the view may be determined with respect to one or more targets of interest in the video environment. In some examples, the view quality score **107** may be generated based on one or more features extracted from the metadata **101**, the video data **105**, and/or the audio data **106**. In some examples, the view quality score **107** may be utilized to compare and/or rank different views or camera angles within the video environment. In some examples, the view quality score **107** may be utilized to determine optimal camera selection, optimal framing, optimal configuration, and/or other optimal control of the one or more video capture device **103** or multiple video capture devices **103** in a multi-camera system.

[0030] The one or more machine learning models **120** may include one or more video machine learning models and/or one or more audio machine learning models to generate at least a portion of the metadata **101**. In some examples, the one or more machine learning models **120** may respectively be configured as a neural network model, a deep learning model, a convolutional neural network model, and/or another type of machine learning models. The metadata **101** provided by the one or more machine learning models **120** may include one or more inferences with respect to the video data **105** and/or the audio data **106**. The metadata engine **122** may perform one or more DSP techniques and/or one or more other prediction techniques that do not involve machine learning to generate at least a portion of the metadata **101**. For example, the metadata engine **122** may determine an average brightness or other quality determinations for a video frame, provide a group eye gaze prediction related to video frames, etc. In another example, the metadata engine **122** may detect motion or proximity of a person in the video environment based on sensor data provided by one or more sensors of the one or more video capture devices **103**. The sensor data may include motion sensor data, proximity sensor data, radar sensor data, LiDAR sensor data, time-of-flight (TOF) sensor data, and/or other sensor data to facilitate detection of a person or object in the video environment. In some examples, the metadata **101** may be related to events with respect to the video data **105** and/or the audio data **106**. In some examples, the metadata **101** may be related to events with respect to raw video data and/or raw audio data. For example, the events may be with respect to video data related to the one or more video capture devices **103**, audio data related to the one or more audio capture devices **102**, sensor data provided by one or more sensors of at least one video capture device of the one or more video capture devices **103**, sensor data provided by at least one audio capture device of the one or more audio capture devices **102**, and/or machine learning model output provided by the one or more machine learning models **120**. In some examples, the raw video data related to the one or more video capture devices **103** may include video data and audio data. Additionally, the metadata **101** may be formatted to assist with view quality scoring and/or control of the one or more video capture devices **103**.

[0031] In some examples, the one or more machine learning models **120** may include one or more machine learning models such as a head pose estimation model that computes a rotational matrix of a detected human face, a face detection model that detects a face in a video frame, an eye gaze estimation model that provides eye tracking with respect to a face in a video frame, a person detection model that detects one or more people in a video frame, an identity detection model that predicts an identity of one or more people in a video frame, an active speaker recognition model that predicts an active speaker in a video frame, an object detection model that detects one or more objects in a video frame, an emotion detection model that predicts a type of emotion related to one or more people in a video frame, a sentiment model that predicts a type of sentiment related to one or more people in a video frame, a noise level prediction model that predicts a degree of noise related to a video frame, a speech detection model that detects speech audio related to a video frame, an audio event model that detects certain types of audio events (e.g., clapping, snapping, whispering, etc.) related to a video frame, a sound source model that classifies a type of sound associated with a sound source related to a video frame, and/or one or more other types of machine learning models. In some examples, a person detected by the person detection model and/or an object detected by the object detection model may be designated for further tracking and/or video processing via one or more future video frames based on audio, sentiment, emotion, interaction events, noise volume, audio classifications, user selection, and/or one or more other criteria related to the person and/or object. In some examples, a person detected by the person detection model and/or an object detected by the object detection model may be designated for further tracking and/or video processing via one or more future video frames based on one or more detected actions with respect to the person and/or object. In a non-limiting example, a whiteboard in a video environment may be designated for further tracking and/or video processing in response to a determination that one or more people are using the whiteboard.

[0032] In some examples, the metadata **101** may include a feature set related to the video data **105** and/or the audio data **106**. One or more features from the feature set may be extracted from one or more video frames related to the video data **105** and/or one or more audio signals related to the audio data **106**. The feature set may include one or more features such as, but not limited to: video features, features related to video frames, object detection features, object classifications, face recognition features, person recognition features, person classifications, three-dimensional coordinates, facial features, mouth features, head pose features, head pose angles, eye features, eye gaze angles, emotion predictions, active speaking classifications, camera locations, camera poses, depth estimation, audio features, and/or one or more other features. In some examples, the feature set is a video feature set that includes one or more features extracted from one or more video frames related to the video data **105**. The video feature set may include visual information related to one or more video frames of the video data **105** such as: object detection information, facial features, motion patterns, color distributions, texture information, and/or other visual features represented by one or more video frames of the video data **105**. In some examples, the video feature set may be generated by applying one or more: image processing techniques, computer vision techniques, and/or machine learning techniques to analyze content of the one or more video frames. In some examples, the feature set is a gaze detection feature set that includes one or more eye features, eye gaze angles, or other gaze detection features provided by the one or more machine learning models **120** and/or the metadata engine **122**.

[0033] In some examples, the video scoring engine **109** generates the view quality score **107** based on respective features included in the metadata **101**. In a non-limiting example, where a head pose estimation model of the one or more machine learning models **120** computes a rotational matrix of a detected human face, the metadata **101** may include a yaw angle that extracted from the rotational matrix. The video scoring engine **109** may compare the yaw angle to one or more other yaw angles for one or more other video capture devices in the video environment to determine the view quality score **107**.

[0034] In some examples, the video scoring engine **109** weights the respective features included in the metadata **101** to determine the view quality score **107**. For example, the view quality score **107** may be a weighted combination of extracted features related to the metadata **101**. Additionally, the video scoring engine **109** may generate the view quality score **107** for each detected face in the video data **105**.

[0035] The weights of the scoring technique utilized by the video scoring engine **109** may depend on a rules set for determining a quality view for a video environment. For example, a weight may correspond to a relative importance of a particular feature included in the metadata **101** after adjusting for differences in scaling of features. The rules set may establish rules related to relationships between features and/or independence of features. For example, the rules set may indicate that nonlinear relationships between features cannot be modeled and/or that one or more features in the metadata **101** are independent with respect to one or more other features in the metadata.

[0036] In some examples, the rules set may establish one or more rules for a particular video environment, a person associated with a digital identifier, and/or a particular object. For example, a first weighting configuration may be beneficial for a first video environment and a second weighting configuration may be beneficial for a second video environment. As such, the rules set may tune weights for a specific use case related to a video environment, a person associated with a digital identifier, and/or a particular object. In a non-limiting example, the rules set may tune weights based on a size of a video environment (e.g., a size of a room) and/or a number of people detected in the video environment. Additionally or alternatively, the rules set may include one or more thresholds and/or rules for thresholds related to different features. In a non-limiting example, the rules set may alter the view quality score **107** if a particular feature such as a head pose feature is greater than a defined threshold value. In some examples, the rules set may disregard a particular weight if the weighting of a feature exceeds a defined threshold value. For instance, if a weight adjustment of a particular feature results in a weighted value that is above a defined threshold value, the rules set may select the original value of the feature prior to the weighting rather than the weighted value.

[0037] In some examples, the view quality score **107** may enable ranking of video capture devices **103** for a particular person or object in the video environment such that a highest ranked video capture device **103** may provide an optimal view with respect to the person or object in the video environment. In some examples, a higher view quality score **107** may correspond to a scenario where a target individual represented in the video data **105** is more frontal (e.g., a head pose angle is closer to zero) and/or more visible with respect to one or more individual or objects represented in the video data **105**.
[0038] The device control engine **110** utilizes the view quality score **107** to determine control data for the one or more video capture devices **103**. In some examples, the device control engine **110** transmits the control data **113** to the one or more video capture devices **103** to configure and/or control the one or more video capture devices **103**. For example, the control data **113** may be utilized to control and/or configure one or more portions of the one or more video capture devices **103**. In some examples, the control data **113** may be additionally utilized to control and/or configure one or more machine learning models of the one or more machine learning models **120**.
[0039] In some examples, the control data **113** may include one or more configuration parameters (e.g., a configuration parameter set) for the one or more video capture devices **103** such as, but not limited to one or more: camera settings, camera selection, camera focus direction, pan, zoom, crop, microphone array settings, beam steering settings, video encoding settings, video frame transmission settings, video frame size, frame rate, color depth settings, resolution format settings, and/or another type of configuration parameter for the one or more video capture devices **103**.
[0040] In some examples, the control data **113** may enable or disable one or more functionalities associated with the one or more video capture devices **103**. For instance, the control data **113** may include one or more control signals and/or configuration data to enable or disable one or more video processing tasks. A video processing task may include camera data acquisition, video encoding/decoding, video machine learning modeling, or another type of video processing task. Additionally, a video processing task may result in generation of metadata, video metrics, object detection, and/or people detection associated with video frames. In some examples, the control data **113** may be additionally or alternatively utilized to: initiate feature extraction with respect to video data, configure parameters or types of features to be extracted, etc.
[0041] The video pipeline engine **111** utilizes the one or more video capture devices **103** configured and/or controlled based on the control data **113** to provide encoded video data **114** related to the one or more video capture devices **103**. In some examples, respective video processors related to the one or more video capture devices **103** are configured and/or control based on the control data **113**. In some examples, respective video processing threads are configured and/or control based on the control data **113**. Configuration of the one or more video capture devices **103** may include: turning particular video processing threads on or off, setting particular parameters for particular video processing threads, initiating particular video related tasks, initiating particular type of encoding task, initiating a video data acquisition task, initiating execution of one a particular machine learning model, and/or one or more other types of configurations for a video processing thread. Based on the configuration of the respective video capture devices **103**, the video pipeline engine **111** may encode video data related to the one or more video capture devices **103** to generate the encoded video data **114**.
[0042] In some examples, the video pipeline engine **111** utilizes the one or more video capture devices **103** configured and/or controlled based on the control data **113** to intelligently switch and/or compose views for a video stream provided by the one or more video capture devices. In some examples, the control data includes a cinematography identifier for a cinematography technique for utilization by the one or more video capture devices **103**. The cinematography identifier may indicate a particular type of cinematography operation to perform with respect to one or more video frames such as, for example, zooming or panning with respect to a person or object within a field of view of a video capture device.

[0043] In some examples, the video pipeline engine **111** outputs the encoded video data **114** to a network device. The network device may be a network switch, a user device, a display device, an edge device, or another type of device communicatively coupled to the video processing system **100** via a network. The network may be a communication network or any suitable network or combination of networks that supports any appropriate protocol suitable for communication of the encoded video data **114** to and from devices. For example, the network may utilize a network communication protocol such as IP, IPoE, or other network communication protocol to transmit the encoded video data **114** via IP datagrams. In some examples, the network may transmit the encoded video data **114** via one or more network layers such as a data link layer. In some examples, the encoded video data **114** may be encapsulated according to a network communication protocol to provide encapsulated video data packets. In some examples, the network is implemented as the Internet, a wireless network, a wired network (e.g., Ethernet), a local area network (LAN), a Wide Area Network (WANs), Bluetooth, Near Field Communication (NFC), or any other type of network that provides communications between one or more components of a network architecture.
[0044] Accordingly, the AV processing system **104** may provide improved video processing as compared to traditional video processing techniques. The AV processing system **104** may additionally or alternatively provide improved audio for the video environment. For example, the encoded video data **114** may be provided with improved accuracy of localization of a sound source in the video environment. The encoded video data **114** may be additionally or alternatively provided with improved audio signals with reduced noise, reverberation, and/or other undesirable audio artifacts even in view of exacting video latency requirements for the encoded video data **114**. For example, the AV processing system **104** may remove or suppress undesirable noise for noise locations in the video environment to provide the encoded video data **114**.
[0045] The AV processing system **104** may also employ fewer computing resources when compared to traditional video processing systems that are used for video processing. Additionally or alternatively, the AV processing system **104** may be configured to deploy a smaller number of memory resources allocated to video processing, beamforming, source separation, denoising, dereverberation, and/or other audio processing for the encoded video data **114**. In some examples, the AV processing system **104** may be configured to improve processing speed of video processing operations, beamforming operations, source separation operations, denoising operations, dereverberation operations, and/or audio filtering operations. These improvements may enable an improved AV processing systems to be deployed with respect to cameras, microphones or other hardware/software configurations where processing and memory resources are limited, and/or where processing speed and efficiency is important.
[0046] FIG. **2** illustrates an example AV processing apparatus **202** configured in accordance with one or more embodiments of the present disclosure. The AV processing apparatus **202** may be configured to perform one or more techniques described in FIG. **1** and/or one or more other techniques described herein.
[0047] The AV processing apparatus **202** may be a computing system communicatively coupled with one or more circuit modules related to video processing and/or audio processing. The AV processing apparatus **202** may comprise or otherwise be in communication with a processor **204**, a memory **206**, video processing circuitry **208**, audio processing circuitry **210**, input/output circuitry **212**, and/or communications circuitry **214**. In some embodiments, the processor **204** (which may comprise multiple or co-processors or any other processing circuitry associated with the processor) may be in communication with the memory **206**.
[0048] The memory **206** may comprise non-transitory memory circuitry and may comprise one or more volatile and/or non-volatile memories. In some examples, the memory **206** may be an electronic storage device (e.g., a computer readable storage medium) configured to store data that may be retrievable by the processor **204**. In some examples, the data stored in the memory **206** may comprise video data, audio data, stereo audio signal data, mono audio signal data, radio frequency

signal data, audio features, video features, control data, machine learning data, defined event type data, or the like, for enabling the AV processing apparatus **202** to carry out various functions or methods in accordance with embodiments of the present disclosure, described herein.

[0049] In some examples, the processor **204** may be embodied in a number of different ways. For example, the processor **204** may be embodied as one or more of various hardware processing means such as a central processing unit (CPU), a microprocessor, a coprocessor, a DSP, a field programmable gate array (FPGA), a neural processing unit (NPU), a graphics processing unit (GPU), a system on chip (SoC), a cloud server processing element, a controller, or a processing element with or without an accompanying DSP. The processor **204** may also be embodied in various other processing circuitry including integrated circuits such as, for example, a microcontroller unit (MCU), an ASIC (application specific integrated circuit), a hardware accelerator, a cloud computing chip, or a special-purpose electronic chip. Furthermore, in some embodiments, the processor **204** may comprise one or more processing cores configured to perform independently. A multi-core processor may enable multiprocessing within a single physical package. Additionally or alternatively, the processor **204** may comprise one or more processors configured in tandem via a bus to enable independent execution of instructions, pipelining, and/or multithreading.

[0050] In some examples, the processor **204** may be configured to execute instructions, such as computer program code or instructions, stored in the memory **206** or otherwise accessible to the processor **204**. Alternatively or additionally, the processor **204** may be configured to execute hard-coded functionality. As such, whether configured by hardware or software instructions, or by a combination thereof, the processor **204** may represent a computing entity (e.g., physically embodied in circuitry) configured to perform operations according to an embodiment of the present disclosure described herein. For example, when the processor **204** is embodied as an CPU, DSP, ARM, FPGA, ASIC, or similar, the processor may be configured as hardware for conducting the operations of an embodiment of the disclosure. Alternatively, when the processor **204** is embodied to execute software or computer program instructions, the instructions may specifically configure the processor **204** to perform the algorithms and/or operations described herein when the instructions are executed. However, in some examples, the processor **204** may be a processor of a device specifically configured to employ an embodiment of the present disclosure by further configuration of the processor using instructions for performing the algorithms and/or operations described herein. The processor **204** may further comprise a clock, an arithmetic logic unit (ALU) and logic gates configured to support operation of the processor **204**, among other things.

[0051] In one or more examples, the AV processing apparatus **202** includes the video processing circuitry **208**. The video processing circuitry **208** may be any means embodied in either hardware or a combination of hardware and software that is configured to perform one or more functions disclosed herein related to the video scoring engine **109**, the device control engine **110**, and/or the video pipeline engine **111**. For example, the video processing circuitry **208** may be any means embodied in either hardware or a combination of hardware and software that is configured to perform one or more functions disclosed herein related to processing of the metadata **101** received from the one or more machine learning models **120** and/or processing of the video data **105** received from the one or more video capture devices **103**. In one or more examples, the AV processing apparatus **202** includes the audio processing circuitry **210**. The audio processing circuitry **210** may be any means embodied in either hardware or a combination of hardware and software that is configured to perform one or more functions disclosed herein related to the audio pipeline engine **112** and/or other audio processing of the audio data **106** received from the one or more audio capture devices **102**.

[0052] In some examples, the AV processing apparatus **202** includes the input/output circuitry **212** that may, in turn, be in communication with processor **204** to provide output to the user and, in some examples, to receive an indication of a user input. The input/output circuitry **212** may comprise a user interface and may comprise a display. In some examples, the input/output circuitry **212** may also comprise a keyboard, a touch screen, touch areas, soft keys, buttons, knobs, or other input/output mechanisms.

[0053] In some examples, the AV processing apparatus **202** includes the communications circuitry **214**. The communications circuitry **214** may be any means embodied in either hardware or a combination of hardware and software that is configured to receive and/or transmit data from/to a network and/or any other device or module in communication with the AV processing apparatus **202**. In this regard, the communications circuitry **214** may comprise, for example, an antenna or one or more other communication devices for enabling communications with a wired or wireless communication network. For example, the communications circuitry **214** may comprise antennae, one or more network interface cards, buses, switches, routers, modems, and supporting hardware and/or software, or any other device suitable for enabling communications via a network. Additionally or alternatively, the communications circuitry **214** may comprise the circuitry for interacting with the antenna/antennae to cause transmission of signals via the antenna/antennae or to handle receipt of signals received via the antenna/antennae.

[0054] FIG. **3** illustrates a network system **300** according to one or more embodiments of the present disclosure. The network system **300** includes the one or more video capture devices **103** (e.g., video capture devices **103***a-n*), a communication center device **302**, and/or a user device **304**. In some examples, at least one of the one or more video capture devices **103** includes the AV processing system **104** and/or the AV processing apparatus **202**. Alternatively, in some examples, the communication center device **302** includes the AV processing system **104** and/or the AV processing apparatus **202**. In some examples, the communication center device **302** includes the AV processing system **104** and/or the AV processing apparatus **202**. The one or more video capture devices **103**, the communication center device **302**, and/or the user device **304** may be communicatively coupled via a network **310**. In some examples, the network **310** includes one or more network devices such as one or more network switches and/or one or more network routers. The communication center device **302** may be a hub device that supports Ethernet, voice over Internet Protocol (VOIP), and/or one or more network communication protocols. In some examples, the communication center device **302** may enable the one or more video capture devices **103** to be configured as a set of network-connected video devices for a video environment.

[0055] The communication center device **302** may provide video and/or audio from the one or more video capture devices **103** to the user device **304**. In some examples, the user device **304** may be configured as a host device for a video conference enabled by the one or more video capture devices **103** and the communication center device **302**. For instance, the user device **304** may be configured as a host of a codec **306** that receives a video stream (e.g., the encoded video data **114**) provided by the one or more video capture device **103**. In some examples, the codec **306** is a video conference codec configured for video conferencing. The user device **304** may be communicatively coupled to the communication center device **302** via the network **410** or another direct IP connection. Alternatively, the user device **304** may be communicatively coupled to the communication center device **302** via a direct wired connection such as a USB connection or another type of hardware interface that supports a display protocol. In some examples, the user device **304** may also be communicatively coupled to the one or more video capture devices **103** via the network **310**. In some examples, the user device **304** may correspond to the communication center device **302** such that the user device **304** manages video and/or audio from the one or more video capture devices **103**. In such examples, the user device **304** includes the AV processing system **104** and/or the AV processing apparatus **202**. In some examples, the user device **304** may additionally or alternatively be configured as a video capture device. As such, video and/or audio from the user device **304** may be provided in addition to video and/or audio from one or more video capture devices **103**.

[0056] The user device **304** may be a smartphone, a laptop, a personal computer, a digital conference device, a wireless conference unit, an augmented reality device, a virtual reality device, or another type of user device. In some examples, the user device **304** includes a display and/or a graphical user interface that renders video content provided by the one or more video capture devices **103**. In some examples, the user device **304** may provide a virtual video capture device and/or a virtual audio capture device for the network system **300**. Additionally, video and/or audio from the virtual devices may be routed to the codec **306** in addition to video and/or audio from one or more video capture devices **103**.

[0057] In some examples, the user device **304** may provide user device data to the communication center device **302** and/or the one or more video capture devices **103** to facilitate interactions with the communication center device **302** and/or the one or more video capture devices **103**. The user device data may include data such as, but not limited to: supported video formats, network interface card (NIC) bandwidth, a role identifier (e.g., hub or video capture device), a device identifier (e.g., a media access control (MAC) address or another type of identifier), a user identifier, and/or other data related to the user device **304**. In some examples, one or more portions of the user device data may be provided via an electronic interface of the user device **304**. Additionally or alternatively, one or more portions of the user device data may be provided via metadata or a user device profile for the user device **304**.

[0058] FIG. **4** illustrates a transmitter system **400** according to one or more embodiments of the present disclosure. The transmitter system **400** may be included in and/or otherwise associated with the one or more video capture devices **103**. In some examples, the video pipeline engine **111** may include the transmitter system **400**. The transmitter system **400** includes a video capture interface **402** and/or an encoder interface **404**. The video capture interface **402** may capture and/or receive one or more portions of video data (e.g., the video data **105**) associated with the one or more video capture devices **103**. The one or more portions of the video data (e.g., the video data **105**) captured and/or received by the video capture interface **402** may be raw video data. In some examples, the video capture interface **402** includes one or more imagers such as one or more camera lenses and/or one or more sensors to capture the video data **105**. In some examples, the video capture interface **402** may utilize a communication protocol and/or a communication connection such as a USB connection, a camera serial interface (CSI) connection, a peripheral component interconnect express (PCIe) connection, an IP connection, or another type of connection to couple the video capture interface **402** to the encoder interface **404**.

[0059] The encoder interface **404** may encode the video data captured and/or received by the video capture interface **402**. In some examples, the encoder interface **404** may transform the video data captured and/or received by the video capture interface **402** into one or more portions of the encoded video data **114**. The encoder interface **404** may also act as an interface between the one or more video capture devices **103** and the network **310**. In some examples, the encoder interface **404** may be configured and/or controlled based on the control data **113**. In some examples, the encoder interface **404** may encode video data via a particular encoding mode (e.g., a first mode for encoding and not transmitting video, a second mode for encoding and transmitting video data, or a third mode for not encoding and not transmitting video data) based on the control data **113**. In some examples, the encoder interface **404** may configure the one or more portions of the encoded video data **114** as video data packets (e.g., IP datagrams) for transmission via the network **310**. For instance, the encoder interface **404** may reformat the video data captured and/or received by the video capture interface **402** into video data IP datagrams by encapsulating the video data IP datagrams via Ethernet frames. In some examples, the encoded video data **114** may include audio data and/or may be synchronized with the audio pipeline engine **112**. In some examples, the one or more machine learning models **120** may extract one or more portions of the metadata **101** from the video data captured and/or received by the video capture interface **402**. In some examples, the control data **113** may be determined based on the metadata **101**.

[0060] FIG. **5** illustrates a receiver system **500** according to one or more embodiments of the present disclosure. The receiver system **500** may be included in and/or otherwise associated with the communication center device **302**. The receiver system **500** includes a video content engine **502** and/or a decoder interface **504**. The video content engine **502** may receive network content from respective video capture devices and/or machine learning models in a video environment. The network content may include metadata, video content, audio content, and/or other content provided by respective video capture devices and/or machine learning models. The video content engine **502** may also provide video data based on the network content received from the content from respective video capture devices. In an example, the video content engine **502** may receive video environment data **501**. In some examples, the video environment data **501** may include the encoded video data **114** provided by the AV processing system **104**. As such, the video content engine **502** may provide the encoded video data **114**. In some examples, the video content engine **502** may provide the encoded video data **114** based on the video environment data **501**. For example, the video content engine **502** may extract a payload that includes the encoded video data **114** from the video environment data **501**.

[0061] In some examples, the video environment data **501** includes multiple captures of a person (e.g., person associated with a digital identifier) from different angles provided by different video capture devices. In some examples, the video environment data **501** additionally or alternatively includes a predicted view quality score (e.g., the view quality score **107**) for each of the viewing angles. As such, the video content engine **502** may determine which viewing angle is an optimal viewing angle such that a corresponding portion of the encoded video data **114** (e.g., corresponding encoded video frames) are provided to the decoder interface **504**.

[0062] The encoded video data **114** may be provided to the decoder interface **504** to transform the encoded video data **114** into decoded video content for rendering via a display interface **506**. In some examples, the decoder interface **504** may determine decoding parameters, frame types, and/or other decoding information for video frames from the encoded video data **114**. The display interface **506** may be a display and/or a graphical user interface of a user device (e.g., the user device **304**).

[0063] FIG. **6** illustrates a score computation architecture **600** according to one or more embodiments of the present disclosure. The score computation architecture **600** may be related to the video scoring engine **109**. In some examples, the metadata **101** includes an object detection feature set **602** and/or a people detection feature set **604**. The object detection feature set **602** may include one or more features related to one or more objects in a video environment associated with the metadata **101**. For example, the object detection feature set **602** may include one or more features related to an object such as, but not limited to: an object classification, three-dimensional coordinates, object detection features, spatial features, motion features, and/or one or more other features. The people detection feature set **604** may include one or more features related to one or more persons of interest in a video environment associated with the metadata **101**. For example, the people detection feature set **604** may include one or more features related to a person such as, but not limited to: face recognition features, person recognition features, person classifications, three-dimensional coordinates, facial features, mouth features, head pose features, head pose angles, eye features, eye gaze angles, emotion predictions, active speaking classifications, and/or one or more other features. In some examples, the object detection feature set **602** may be formatted as an object feature vector and/or the people detection feature set **604** may be formatted as an identity feature vector.

[0064] In some examples, the video scoring engine **109** may generate an object view score **107***a* based on the object detection feature set **602**. The object view score **107***a* may refer to a numerical or categorical value that represents quality of a view for an object captured by the one or more video capture device **103**. For example, the object view score **107***a* may be determined with respect to an object in the video environment. Additionally or alternatively, the video scoring engine **109** may generate a person view score **107***b* based on the object detection feature set **602**. The person view score **107***b* may refer to a numerical or categorical value that represents quality of a view for a person captured by the one or more video capture device **103**. For example, the person view score **107***b* may be determined with respect to a person in the video environment. In some examples, the view quality score **107** includes and/or is determined based on the object view score **107***a* and/or the person view score **107***b*. In some examples, the video scoring engine **109** generates the object view score **107***a* based on respective features included in the object detection feature set **602**. In some examples, the video scoring engine **109** weights the respective features included in the object detection feature set **602** to determine the object view score **107***a*. For example, the object view score **107***a* may be a weighted combination of extracted features related to the object detection feature set **602**. In some examples, the object view score **107***a* and/or the person view score **107***b* may be based on a size of a detection area for a corresponding object or person. For example, the object view score **107***a* and/or the person view score **107***b* may be weighted based on an area of a bounding box in a video frame for a corresponding object or person. In some examples, a weight associated with a bounding box may scale a size value of the bounding box. In some examples, a view score for a person or object may include the scaled size value of the bounding box. In some examples, a weight associated with a bounding box may be based on parameters for the video environment such as, but not limited to, a size of the video environment, a number of people detected in the video environment, a type of object or person associated with the bounding box, etc.

[0065] In some examples, the video scoring engine **109** generates the person view score **107***b* based on respective features included in the people detection feature set **604**. In some examples, the video scoring engine **109** weights the respective features included in the people detection feature set **604** to determine the person view score **107***b*. For example, the person view score **107***b* may be a weighted combination of extracted features related to the people detection feature set **604**. In some examples, the video scoring engine **109** may correlate the object view score **107***a* and/or the person view score **107***b* with entity data **606** related to an entity of interest in one or more video frames. In some examples, the video scoring engine **109** may store the entity data **606** in a data storage **608**. The data storage **608** may correspond to the memory **206** or another storage communicatively coupled to the video scoring engine **109**. In some examples, the device control engine **110** may utilize the entity data **606** to generate one or more portions of the control data **113** to focus on and/or determine a best view for the object and/or the entity of interest. In some examples, the video scoring engine **109** may store the entity data **606** in a data storage **608** based on 3D coordinates, object classifications, and/or person classifications related to the entity data **606**. For example, the video scoring engine **109** may update and/or create data for a particular 3D coordinate, object classification, and/or person classification in a video environment based on the 3D coordinates, object classifications, and/or person classifications related to the entity data **606**.

[0066] FIG. **7** illustrates a device control architecture **700** according to one or more embodiments of the present disclosure. The device control architecture **700** may be related to the device control engine **110**. In some examples, the device control engine **110** utilizes the entity data **606** stored in the data storage **608** to generate one or more portions of the control data **113**. For example, the device control engine **110** may utilize a group eye gaze estimation and/or an active speaker selection associated with the entity data **606** to generate one or more portions of the control data **113**. The group eye gaze estimation may infer where a person in a video environment is looking. For example, if a person is looking in a direction of a particular detected object such as a whiteboard in the video environment, the whiteboard may be selected as a focus target. In some examples, the group eye gaze estimation may be computed based on features such as eye gaze angle, 3D coordinates, etc. The active speaker selection may select a focus target predicted to be an active speaker in the video environment. In some examples, the active speaker selection may be computed based on an active speaking classification.

[0067] In some examples, the control data **113** includes device selection data **702** and/or device configuration data **704**. The device selection data **702** may include a video capture device identifier that corresponds to a video capture device **103** to be selected for operation in the video environment. The device configuration data **704** may include one or more configuration parameters for one or more video capture devices **103**. For example, the device configuration data **704** may include one or more: camera settings, exposure, white balance, color temperature, camera selection, camera mode selection, camera focus direction, pan, zoom, crop, microphone selection, microphone array settings, beam steering settings, speech separation, video encoding settings, video frame transmission settings, video frame size, frame rate, color depth settings, resolution format settings, machine learning model settings, metadata selection settings, optical character recognition (OCR) settings, and/or another type of configuration parameter for the one or more video capture devices **103**. In some examples, the device selection data **702** may select an optimal video capture device **103** for a focus target. In some examples, the device configuration data **704** may control a video capture device **103** and/or a related video stream from a selected video capture device identified in the device selection data **702**.

[0068] FIG. **8** illustrates a machine learning architecture **800** according to one or more embodiments of the present disclosure. The machine learning architecture **800** may be related to the one or more machine learning models **120**. In some examples, the one or more machine learning models **120** may provide one or more features **802** extracted from one or more video frames of the video data **105**. The one or more features **802** may be related to object detection, depth estimation, people detection, entity localization, photogrammetry, camera calibration, head pose estimation, eye gaze estimation, facial landmark detection, identity feature extraction, emotion recognition, active speak recognition, and/or another type of machine learning feature extraction technique. In some examples, the one or more features **802** may be utilized to generate at least a portion of the object detection feature set **602** and/or the people detection feature set **604** for the one or more video frames. Additionally, the metadata **101** may include the one or more features **802**, the object detection feature set **602**, the people detection feature set **604**, and/or one or more other features related to the video data **105** and/or the audio data **106**. In some examples, the machine learning architecture **800** may be executed via an edge devices that captures the one or video frames and/or other sensor input related to the video data **105** and/or the audio data **106**.

[0069] In some examples, one or more features extracted from one or more video frames provided by a first video capture device may be correlated with one or more other features extracted from one or more other video frames provided by a second video capture device. The features may be correlated, for example, based on information provided by a calibration and/or photogrammetry process for the first video capture device and the second video capture device. For example, the first video capture device and the second video capture device may be calibrated and/or configured for the video environment via a photogrammetry process such that camera location information (e.g., x, y and z coordinates), pose information, and/or a projection matrix for the first video capture device and the second video capture device are determined. In some examples, the camera location information, pose information, and/or projection matrix may be utilized for entity localization, feature extraction, and/or one or more other types of modeling by the one or more machine learning models **120**.

[0070] In some examples, photogrammetry and/or a related photogrammetry feature set may be utilized to estimate a position of a person or object in the video environment. For example, the photogrammetry feature set may refer to collection of data or measurements derived from analyzing multiple images or video frames to reconstruct three-dimensional information regarding objects or scenes in a video environment. The photogrammetry feature set may include: spatial coordinates, camera positions, orientation parameters, depth maps, geometric information, and/or other photogrammetry information extracted from images captured from different viewpoints. In some examples, a set of overlapping images of the video environment may be captured during calibration to generate a projection matrix associated with the video environment. The projection matrix may map 3D coordinates of the video environment to 2D camera coordinates associated with video capture devices in the video environment. In some examples, an inverse of the projection matrix associated with camera position may be utilized to determine a 3D vector associated with a particular space in the video environment and a particular pixel of an image. For example, given a coordinate of a bounding box of a detected person or object, a vector through the 3D space of the video environment may be determined such that the detected person or object is deemed to be located along the vector. In some examples, a depth along the vector may be estimated based on triangulation associated with an intersection of vectors across video capture devices in a 3D space. Additionally or alternatively, a depth along the vector may be estimated based on sensor data, depth associated with a video signal, and/or machine learning model output.

[0071] FIG. **9** illustrates an example video environment **902** according to one or more embodiments of the present disclosure. The video environment **902** may be an indoor environment, an outdoor environment, an entertainment environment, a room, a conference room, a meeting room, a classroom, a lecture hall, a performance hall, a broadcasting environment, a sports stadium or arena, a virtual environment, an automobile environment, or another type of video environment. The video environment **902** includes at least the one or more video capture devices **103***a-n* that are respectively capable of capturing video and/or audio from one or more sources and/or other audio in the video environment **902**. For example, the one or more video capture devices **103***a-n* **102** may capture video and/or audio (e.g., the video data **105** and/or the audio data **106**) associated with a target talker **904**, a target object **905**, undesirable speech **906**, and/or noise **908** in the video environment **902**. In some examples, the one or more video capture devices **103***a-n* recognize the target talker **904**, modifies a video capture process, and/or steers one or more audio beams based on the control data **113**.

[0072] Embodiments of the present disclosure are described below with reference to block diagrams and flowchart illustrations. Thus, it should be understood that each block of the block diagrams and flowchart illustrations may be implemented in the form of a computer program product, an entirely hardware embodiment, a combination of hardware and computer program products, and/or apparatus, systems, computing devices/entities, computing entities, and/or the like carrying out instructions, operations, steps, and similar words used interchangeably (e.g., the executable instructions, instructions for execution, program code, and/or the like) on a computer-readable storage medium for execution. For example, retrieval, loading, and execution of code may be performed sequentially such that one instruction is retrieved, loaded, and executed at a time.

[0073] In some example embodiments, retrieval, loading, and/or execution may be performed in parallel such that multiple instructions are retrieved, loaded, and/or executed together. Thus, such embodiments may produce specifically-configured machines performing the steps or operations specified in the block diagrams and flowchart illustrations. Accordingly, the block diagrams and flowchart illustrations support various combinations of embodiments for performing the specified instructions, operations, or steps.

[0074] FIG. **10** is a flowchart diagram of an example process **1000** for providing video capture device control based on metadata related to a video environment, in accordance with, for example, the AV processing apparatus **202** illustrated in FIG. **2**. Via the various operations of the process **1000**, the AV processing apparatus **202** may enhance quality, reliability, and/or source separation of video data for rendering via a display interface.

[0075] The process **1000** begins at operation **1002** that receives (e.g., by the video processing circuitry **208** and/or the audio processing circuitry **210**) metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment. The video environment may be an indoor environment, an outdoor environment, an entertainment environment, a room, a conference room, a meeting room, a classroom, a lecture hall, a performance hall, a broadcasting environment, a sports stadium or arena, a virtual environment, an automobile environment, or another type of video environment. The metadata may include one or more inferences with respect to the video data and/or the audio data. In some examples, the metadata may include a feature set associated with the video data and/or the audio data. For example, the feature set may include one or more features such as, but not limited to: video features, features associated with video frames, object detection features, object classifications, face recognition features, person recognition features, person classifications, three-dimensional coordinates, facial features, mouth features, head pose features, head pose angles, eye features, eye gaze angles, emotion predictions, active speaking classifications, camera locations, camera poses, depth estimation, color format, frame orientation, frame rotation, natural language processing, video quality, audio features, and/or one or more other features. In some examples, depth estimation metadata may include a depth estimation feature set provided by the at least one machine learning model. In some examples, the depth estimation feature set may include: depth map features, point cloud features, and/or depth estimation features associated with a three-dimensional structure derived from two-dimensional visual input. Additionally or alternatively, depth estimation metadata may include depth estimation data associated with stereo camera data, sensor data (e.g., sensor data associated with a TOF sensor, a LiDAR sensor, and/or another type of sensor), photogrammetry data, and/or other data. In some examples, color format metadata may indicate a type of color format (e.g., RGB, YUV, NV12, etc.) associated with video frames. In some examples, frame orientation metadata may indicate whether video frames are oriented as a landscape orientation or a portrait orientation. In some examples, frame rotation metadata may indicate whether a video frame is rotated horizontally, vertically, diagonally, etc.

[0076] The process **1000** also includes an operation **1004** that generates (e.g., by the video processing circuitry **208**) a view quality score for the video environment based at least in part on the metadata. For example, the view quality score may be based on respective features included in the metadata. In some examples, the view quality score may be a weighted combination of extracted features associated with the metadata.

[0077] The process **1000** also includes an operation **1006** that generates (e.g., by the video processing circuitry **208**) control data for the at least one video capture device based at least in part on the view quality score. The control data may be utilized to control and/or configure the at least one video capture device. Control and/or configuration of the at least one video capture device may include: turning particular video processing threads on or off, setting particular parameters for particular video processing threads, initiating particular video related tasks, initiating particular type of encoding task, initiating a video data acquisition task, initiating execution of one a particular machine learning model, enabling speech separation with respect to a video processing thread, modifying one or more video frames associated with a video processing thread, enabling an OCR task associated with one or more video frames associated with a video processing thread, and/or one or more other types of configurations for a video processing thread. In some examples, modifying one or more video frames associated with a video processing thread includes enabling a digital zoom, pan, and/or crop associated with a particular region in one or more video frames associated with a video processing thread. The particular region may include a detected person, a person associated with a digital identifier, a person associated with speech separation, a particular object, a whiteboard region, etc.

[0078] The process **1000** also includes an operation **1008** that outputs (e.g., by the input/output circuitry **212**) the control data to the at least one video capture device.

[0079] In some examples, the metadata is first metadata. In some examples, second metadata generated via digital signal processing associated with a metadata engine that is different than the at least one machine learning model is received. In some examples, the view quality score for the video environment is generated based on the first metadata and the second metadata.

[0080] In some examples, a video feature set provided by the at least one machine learning model is received. In some examples, the view quality score for the video environment is generated based on the video feature set.

[0081] In some examples, the view quality score for the video environment is modified based on

respective weights for respective features included in the video feature set.
[0082] In some examples, an object detection feature set provided by the at least one machine learning model is received. In some examples, the view quality score for the video environment is generated based on the object detection feature set.
[0083] In some examples, a people detection feature set provided by the at least one machine learning model is received. In some examples, the view quality score for the video environment is generated based on the people detection feature set.
[0084] In some examples, a gaze detection feature set provided by the at least one machine learning model is received. In some examples, the view quality score for the video environment is generated based on the gaze detection feature set.
[0085] In some examples, the view quality score for the video environment is generated based on a photogrammetry feature set associated with the at least one video capture device.
[0086] In some examples, the view quality score for the video environment is generated based on depth estimation metadata.
[0087] In some examples, a depth estimation feature set provided by the at least one machine learning model is received. In some examples, the view quality score for the video environment is generated based on the depth estimation feature set.
[0088] In some examples, a configuration parameter set for the at least one video capture device is generated based on the view quality score. In some examples, the configuration parameter set is output to the at least one video capture device.
[0089] In some examples, device selection data for the at least one video capture device is generated based on the view quality score. In some examples, the device selection data is output to the at least one video capture device.
[0090] In some examples, a microphone array beam for an audio capture device in the video environment is steered based on the control data.
[0091] In some examples, an object view score for an object in the video environment is generated based on the metadata. In some examples, the control data for the at least one video capture device is generated based on the object view score.
[0092] In some examples, a person view score for a person in the video environment is generated based on the metadata. In some examples, the control data for the at least one video capture device is generated based on the person view score.
[0093] In some examples, one or more video frames associated with the at least one video capture device are output based on the control data.
[0094] FIG. **11** is a flowchart diagram of an example process **1100** for providing video capture device control based on metadata related to a video environment, in accordance with, for example, the AV processing apparatus **202** illustrated in FIG. **2**. Via the various operations of the process **1100**, the AV processing apparatus **202** may enhance quality, reliability, and/or source separation of video data for rendering via a display interface.
[0095] The process **1100** begins at operation **1102** that receives (e.g., by the video processing circuitry **208** and/or the audio processing circuitry **210**) metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment. The video environment may be an indoor environment, an outdoor environment, an entertainment environment, a room, a conference room, a meeting room, a classroom, a lecture hall, a performance hall, a broadcasting environment, a sports stadium or arena, a virtual environment, an automobile environment, or another type of video environment. The metadata may include one or more inferences with respect to the video data and/or the audio data. In some examples, the metadata may include a feature set associated with the video data and/or the audio data. For example, the feature set may include one or more features such as, but not limited to: video features, features associated with video frames, object detection features, object classifications, face recognition features, person recognition features, person classifications, three-

dimensional coordinates, facial features, mouth features, head pose features, head pose angles, eye features, eye gaze angles, emotion predictions, active speaking classifications, camera locations, camera poses, depth estimation, color format, frame orientation, frame rotation, natural language processing, video quality, audio features, and/or one or more other features. In some examples, depth estimation metadata may include a depth estimation feature set provided by the at least one machine learning model. Additionally or alternatively, depth estimation metadata may include depth estimation data associated with stereo camera data, sensor data (e.g., sensor data associated with a TOF sensor, a LiDAR sensor, and/or another type of sensor), photogrammetry data, and/or other data. In some examples, color format metadata may indicate a type of color format (e.g., RGB, YUV, NV12, etc.) associated with video frames. In some examples, frame orientation metadata may indicate whether video frames are oriented as a landscape orientation or a portrait orientation. In some examples, frame rotation metadata may indicate whether a video frame is rotated horizontally, vertically, diagonally, etc.

[0096] The process **1100** also includes an operation **1104** that generates (e.g., by the video processing circuitry **208**) a view quality score for the video environment based at least in part on the metadata. For example, the view quality score may be based on respective features included in the metadata. In some examples, the view quality score may be a weighted combination of extracted features associated with the metadata.

[0097] The process **1100** also includes an operation **1106** that outputs (e.g., by the input/output circuitry **212**) the view quality score to a network device.

[0098] The process **1100** also includes an operation **1108** that receives (e.g., by the input/output circuitry **212**) control data for the at least one video capture device based at least in part on the view quality score.

[0099] The process **1100** begins at operation **1110** that configures (e.g., by the video processing circuitry **208** and/or the audio processing circuitry **210**) the at least one video capture device based at least in part on the control data. Control and/or configuration of the at least one video capture device may include: turning particular video processing threads on or off, setting particular parameters for particular video processing threads, initiating particular video related tasks, initiating particular type of encoding task, initiating a video data acquisition task, initiating execution of one a particular machine learning model, enabling speech separation with respect to a video processing thread, modifying one or more video frames associated with a video processing thread, enabling an OCR task associated with one or more video frames associated with a video processing thread, and/or one or more other types of configurations for a video processing thread. In some examples, modifying one or more video frames associated with a video processing thread includes enabling a digital zoom, pan, and/or crop associated with a particular region in one or more video frames associated with a video processing thread. The particular region may include a detected person, a person associated with a digital identifier, a person associated with speech separation, a particular object, a whiteboard region, etc.

[0100] In some examples, one or more video frames associated with the at least one video capture device may be output based on the control data.

[0101] In some examples, video data associated with the at least one video capture device may be encoded based on the control data. In some examples, the encoded video data may be output to the network device.

[0102] In some examples, one or more video frames associated with the at least one video capture device may be output based on a determination that one or more features included in the metadata satisfies threshold value criteria.

[0103] Although example processing systems have been described in the figures herein, implementations of the subject matter and the functional operations described herein may be implemented in other types of digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them.

[0104] Embodiments of the subject matter and the operations described herein may be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the subject matter described herein may be implemented as one or more computer programs, i.e., one or more modules of computer program instructions, encoded on computer-readable storage medium for execution by, or to control the operation of, information/data processing apparatus. Alternatively, or in addition, the program instructions may be encoded on an artificially-generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal, which is generated to encode information/data for transmission to suitable receiver apparatus for execution by an information/data processing apparatus. A computer-readable storage medium may be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. Moreover, while a computer-readable storage medium is not a propagated signal, a computer-readable storage medium may be a source or destination of computer program instructions encoded in an artificially-generated propagated signal. The computer-readable storage medium may also be, or be included in, one or more separate physical components or media (e.g., multiple CDs, disks, or other storage devices).

[0105] A computer program (also known as a program, software, software application, script, or code) may be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it may be deployed in any form, including as a stand-alone program or as a module, engine, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program may, but need not, correspond to a file in a file system. A program may be stored in a portion of a file that holds other programs or information/data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub-programs, or portions of code). A computer program may be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

[0106] The processes and logic flows described herein may be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input information/data and generating output. Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and information/data from a read-only memory, a random access memory, or both. The essential elements of a computer are a processor for performing actions in accordance with instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive information/data from or transfer information/data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto-optical disks, or optical disks. However, a computer need not have such devices. Devices suitable for storing computer program instructions and information/data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory may be supplemented by, or incorporated in, special purpose logic circuitry.

[0107] The term “or” is used herein in both the alternative and conjunctive sense, unless otherwise indicated. The terms “illustrative,” “example,” and “exemplary” are used to be examples with no indication of quality level. Like numbers refer to like elements throughout.

[0108] The term "comprising" means "including but not limited to," and should be interpreted in the manner it is typically used in the patent context. Use of broader terms such as comprises, includes, and having should be understood to provide support for narrower terms, such as consisting of, consisting essentially of, comprised substantially of, and/or the like.
[0109] The phrases "in one embodiment," "according to one embodiment," and the like generally mean that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure, and may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).
[0110] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any disclosures or of what may be claimed, but rather as description of features specific to particular embodiments of particular disclosures. Certain features that are described herein in the context of separate embodiments may also be implemented in combination in a single embodiment.
[0111] Conversely, various features that are described in the context of a single embodiment may also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.
[0112] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in incremental order, or that all illustrated operations be performed, to achieve desirable results, unless described otherwise. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems may generally be integrated together in a product or packaged into multiple products.
[0113] Thus, particular embodiments of the subject matter have been described. Other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims may be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or incremental order, to achieve desirable results, unless described otherwise. In certain implementations, multitasking and parallel processing may be advantageous.
[0114] Hereinafter, various characteristics will be highlighted in a set of numbered clauses or paragraphs. These characteristics are not to be interpreted as being limiting on the disclosure or inventive concept, but are provided merely as a highlighting of some characteristics as described herein, without suggesting a particular order of importance or relevancy of such characteristics.
[0115] Clause 1. An apparatus comprising at least one processor and a memory storing instructions that are operable, when executed by the processor, to cause the apparatus to: receive metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment.
[0116] Clause 2. The apparatus of clause 1, wherein the instructions are further operable to cause the apparatus to: generate a view quality score for the video environment based at least in part on the metadata.
[0117] Clause 3. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate control data for the at least one video capture device based at least in part on the view quality score.
[0118] Clause 4. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: output the control data to the at least one video capture

device.
[0119] Clause 5. The apparatus of any one of the foregoing clauses, wherein the metadata is first metadata and the instructions are further operable to cause the apparatus to: receive second metadata generated via digital signal processing associated with a metadata engine that is different than the at least one machine learning model.
[0120] Clause 6. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on the first metadata and the second metadata.
[0121] Clause 7. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: receive a video feature set provided by the at least one machine learning model.
[0122] Clause 8. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on the video feature set.
[0123] Clause 9. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: modify the view quality score for the video environment based at least in part on respective weights for respective features included in the video feature set.
[0124] Clause 10. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: receive an object detection feature set provided by the at least one machine learning model.
[0125] Clause 11. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on the object detection feature set.
[0126] Clause 12. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: receive a people detection feature set provided by the at least one machine learning model.
[0127] Clause 13. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on the people detection feature set.
[0128] Clause 14. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: receive a gaze detection feature set provided by the at least one machine learning model.
[0129] Clause 15. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on the gaze detection feature set.
[0130] Clause 16. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on a photogrammetry feature set associated with the at least one video capture device.
[0131] Clause 17. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on depth estimation metadata.
[0132] Clause 18. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: receive a depth estimation feature set provided by the at least one machine learning model.
[0133] Clause 19. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on the depth estimation feature set.
[0134] Clause 20. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate a configuration parameter set for the at least

one video capture device based at least in part on the view quality score.
[0135] Clause 21. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: output the configuration parameter set to the at least one video capture device.
[0136] Clause 22. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate device selection data for the at least one video capture device based at least in part on the view quality score.
[0137] Clause 23. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: output the device selection data to the at least one video capture device.
[0138] Clause 24. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: steer a microphone array beam for an audio capture device in the video environment based at least in part on the control data.
[0139] Clause 25. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate an object view score for an object in the video environment based at least in part on the metadata.
[0140] Clause 26. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the control data for the at least one video capture device based at least in part on the object view score.
[0141] Clause 27. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate person view score for a person in the video environment based at least in part on the metadata.
[0142] Clause 28. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: generate the control data for the at least one video capture device based at least in part on the person view score.
[0143] Clause 29. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: output one or more video frames related to the at least one video capture device based at least in part on the control data.
[0144] Clause 30. A computer-implemented method comprising steps in accordance with any one of the foregoing clauses.
[0145] Clause 31. A computer program product, stored on a computer readable medium, comprising instructions that, when executed by one or more processors of the audio signal processing apparatus, cause the one or more processors to perform one or more operations related to any one of the foregoing clauses.
[0146] Clause 32. An apparatus comprising at least one processor and a memory storing instructions that are operable, when executed by the processor, to cause the apparatus to: generate metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment.
[0147] Clause 33. The apparatus of clause 32, wherein the instructions are further operable to cause the apparatus to: generate a view quality score for the video environment based at least in part on the metadata.
[0148] Clause 34. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: output the view quality score to a network device.
[0149] Clause 35. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: receive control data for the at least one video capture device based at least in part on the view quality score.
[0150] Clause 36. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: configure the at least one video capture device based at least in part on the control data.
[0151] Clause 37. The apparatus of any one of the foregoing clauses, wherein the instructions are

further operable to cause the apparatus to: output one or more video frames associated with the at least one video capture device based at least in part on the control data.
[0152] Clause 38. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: encode video data associated with the at least one video capture device based at least in part on the control data.
[0153] Clause 39. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: output the encoded video data to the network device.
[0154] Clause 40. The apparatus of any one of the foregoing clauses, wherein the instructions are further operable to cause the apparatus to: output one or more video frames associated with the at least one video capture device based on a determination that one or more features included in the metadata satisfies threshold value criteria.
[0155] Clause 41. A computer-implemented method comprising steps in accordance with any one of the foregoing clauses.
[0156] Clause 42. A computer program product, stored on a computer readable medium, comprising instructions that, when executed by one or more processors of the audio signal processing apparatus, cause the one or more processors to perform one or more operations related to any one of the foregoing clauses.
[0157] Many modifications and other embodiments of the disclosures set forth herein will come to mind to one skilled in the art to which these disclosures pertain having the benefit of the teachings presented in the foregoing description and the associated drawings. Therefore, it is to be understood that the disclosures are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation, unless described otherwise.

Claims

1. An apparatus comprising at least one processor and a memory storing instructions that are operable, when executed by the processor, to cause the apparatus to: receive metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment; generate a view quality score for the video environment based at least in part on the metadata; generate control data for the at least one video capture device based at least in part on the view quality score; and output the control data to the at least one video capture device.
2. The apparatus of claim 1, wherein the metadata is first metadata, and wherein the instructions are further operable to cause the apparatus to: receive second metadata generated via digital signal processing associated with a metadata engine that is different than the at least one machine learning model; and generate the view quality score for the video environment based at least in part on the first metadata and the second metadata.
3. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: receive a video feature set provided by the at least one machine learning model; and generate the view quality score for the video environment based at least in part on the video feature set.
4. The apparatus of claim 3, wherein the instructions are further operable to cause the apparatus to: modify the view quality score for the video environment based at least in part on respective weights for respective features included in the video feature set.
5. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: receive an object detection feature set provided by the at least one machine learning model; and generate the view quality score for the video environment based at least in part on the object detection feature set.
6. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to:

receive a people detection feature set provided by the at least one machine learning model; and generate the view quality score for the video environment based at least in part on the people detection feature set.

7. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: receive a gaze detection feature set provided by the at least one machine learning model; and generate the view quality score for the video environment based at least in part on the gaze detection feature set.

8. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on a photogrammetry feature set associated with the at least one video capture device.

9. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: generate the view quality score for the video environment based at least in part on depth estimation metadata.

10. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: receive a depth estimation feature set provided by the at least one machine learning model; and generate the view quality score for the video environment based at least in part on the depth estimation feature set.

11. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: generate a configuration parameter set for the at least one video capture device based at least in part on the view quality score; and output the configuration parameter set to the at least one video capture device.

12. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: generate device selection data for the at least one video capture device based at least in part on the view quality score; and output the device selection data to the at least one video capture device.

13. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: steer a microphone array beam for an audio capture device in the video environment based at least in part on the control data.

14. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: generate an object view score for an object in the video environment based at least in part on the metadata; and generate the control data for the at least one video capture device based at least in part on the object view score.

15. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: generate person view score for a person in the video environment based at least in part on the metadata; and generate the control data for the at least one video capture device based at least in part on the person view score.

16. The apparatus of claim 1, wherein the instructions are further operable to cause the apparatus to: output one or more video frames associated with the at least one video capture device based at least in part on the control data.

17. A computer-implemented method comprising: receiving metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment; generating a view quality score for the video environment based at least in part on the metadata; generating control data for the at least one video capture device based at least in part on the view quality score; and outputting the control data to the at least one video capture device.

18. The computer-implemented method of claim 17, wherein the metadata is first metadata, and the computer-implemented method further comprising: receiving second metadata generated via digital signal processing associated with a metadata engine that is different than the at least one machine learning model; and generating the view quality score for the video environment based at least in part on the first metadata and the second metadata.

19. The computer-implemented method of claim 17, further comprising: receiving a video feature set provided by the at least one machine learning model; and generating the view quality score for the video environment based at least in part on the video feature set.

20. A computer program product, stored on a computer readable medium, comprising instructions that, when executed by one or more processors of an apparatus, cause the one or more processors to: receive metadata generated by at least one machine learning model associated with at least one video capture device located within a video environment; generate a view quality score for the video environment based at least in part on the metadata; generate control data for the at least one video capture device based at least in part on the view quality score; and output the control data to the at least one video capture device.